

Applying IEC 62443 to Data Center Criticality: A Zone and Conduit Model Based on Uptime Tiers

The New Critical Infrastructure: IT/OT Convergence in the Modern Data Center

Introduction: The Data Center as a Cyber-Physical System

The modern data center is a highly complex cyber-physical system, an environment where Information Technology (IT) and Operational Technology (OT) are not merely collocated but deeply converged and interdependent. The function of the IT infrastructure—the servers, storage, and networks that process data—is entirely dependent on the continuous, correct operation of the underlying OT infrastructure: the power, cooling, and physical security systems. Historically, these OT systems, such as Building Management Systems (BMS), Electrical Power Monitoring Systems (EPMS), and chiller plants, were proprietary, standalone systems isolated by an "air gap". This is no longer the case. The drive for efficiency, remote monitoring, predictive maintenance, and sustainability has led to the widespread integration of these legacy systems with IP-based networks. This convergence, managed through platforms like Data Center Infrastructure Management (DCIM), has connected previously isolated control systems to the enterprise network and, in some cases, the internet.

This IT/OT convergence has created a new, dangerously expanded attack surface. A cyberattack is no longer limited to stealing data from a server (an IT threat). An attacker can now target the facility's OT, such as its power distribution units (PDUs) or HVAC controllers. Recent threat intelligence confirms that advanced adversaries are actively researching and developing tactics to target these data center OT systems. The consequence of a successful OT attack—such as shutting down a chiller plant or tripping a UPS—is not data loss, but catastrophic physical downtime, permanent equipment damage, and a complete violation of the data center's core business promise: availability. This fundamental shift in risk—from protecting data *confidentiality* to ensuring *physical availability*—mandates a new security framework.

The "Previous Pattern": Limitations of the Purdue Model in a Data Center Context

The "previous pattern" for Industrial Automation and Control System (IACS) network segmentation is the Purdue Enterprise Reference Architecture (PERA), or Purdue Model. This model defines a rigid, hierarchical structure of five levels, separating the enterprise IT network (Level 4/5) from the industrial control and physical processes (Levels 0-3) via a demilitarized zone (DMZ).

This model is increasingly recognized as obsolete for modern industrial environments, and it is particularly ill-suited for a data center.

- **Rigid Hierarchy:** The Purdue Model's strict hierarchy fails to represent the data flows of a modern facility. For example, a "Level 1" smart sensor on a chiller (OT) may need to send data directly to a "Level 5" cloud analytics platform (IT) for predictive maintenance,

- bypassing the prescribed levels.
- **Blurred Boundaries:** A data center *is* the convergence. The enterprise network (Level 4/5) and the industrial control systems (Level 1-3) are not in separate facilities; they are in the same building, often managed by the same organization.
- **False Sense of Security:** The model's foundational "air-gap" mentality is no longer a reality. The IP-based connectivity of modern BMS and EPMS systems renders the air-gap concept invalid.

The IEC 62443 standard provides the necessary evolution. It replaces the rigid, hierarchical *blueprint* of the Purdue Model with a flexible, risk-based *philosophy*. The core concept of IEC 62443 is to group assets not by their hierarchical level, but by their *common security requirements*. This methodology, known as defining Zones and Conduits, is uniquely capable of segmenting the complex, non-linear data flows of a modern, converged data center.

Bridging Resilience Frameworks: Uptime Tiers and IEC 62443 Security Levels

To build a relevant IEC 62443 model for a data center, two frameworks must be integrated: the Uptime Institute's Tier Standard, which defines *criticality*, and IEC 62443's Security Levels, which define *security*.

Framework 1: Uptime Institute Tiers as a Consequence Analysis

The Uptime Institute's Tier Classification System is the global standard for defining data center reliability and criticality. The four Tiers define specific requirements for redundancy, maintainability, and fault tolerance.

- **Tier I: Basic Capacity:** Single path for power and cooling; no redundancy.
- **Tier II: Redundant Capacity:** Adds redundant components (N+1), but maintains a single path for distribution.
- **Tier III: Concurrently Maintainable:** Multiple power and cooling distribution paths (N+1), allowing any component to be taken offline for maintenance without impacting the IT load.
- **Tier IV: Fault Tolerant:** Multiple, independent, active distribution paths (2N or 2N+1); automatically withstands any *single* equipment failure or distribution path interruption.

These Tiers are not just design specifications; they represent a pre-calculated *business impact and consequence analysis*. The financial investment and availability guarantees (ranging from 99.671% for Tier I to 99.995% for Tier IV) are a direct proxy for the organization's tolerance for failure and, by extension, the *consequence* of that failure.

This consequence analysis is the primary input for the IEC 62443-3-2 risk assessment process. This standard mandates that the Target Security Level (SL-T) for a zone be determined by assessing the *potential consequences* of a compromise within that zone. Therefore, the Uptime Tier of a facility is the most logical and essential starting point for determining the SL-T of its critical OT zones.

Table 1: Uptime Tier Characteristics and Cyber-Physical Implications

Tier Level	Uptime Guarantee	Redundancy Model	Key Feature	Cyber-Physical Implication & Consequence
Tier I	99.671% (<28.8 hrs/yr)	N	Basic Capacity	Consequence: Low. Downtime is expected. Implication: Basic

Tier Level	Uptime Guarantee	Redundancy Model	Key Feature	Cyber-Physical Implication & Consequence
				cyber controls are sufficient.
Tier II	99.741% (<22 hrs/yr)	N+1 Components	Redundant Components	Consequence: Moderate. Unplanned downtime is disruptive. Implication: Must protect against simple, intentional attacks.
Tier III	99.982% (<1.6 hrs/yr)	N+1 Paths	Concurrently Maintainable	Consequence: High. Downtime is a major business event. Implication: Must be resilient to planned (maintenance) events and <i>sophisticated</i> attacks.
Tier IV	99.995% (<26.3 min/yr)	2N Paths	Fault Tolerant	Consequence: Catastrophic. Downtime is unacceptable. Implication: Must withstand <i>any</i> single failure, including cyber-induced failures.

Framework 2: IEC 62443 Security Levels (SLs) as the Security Metric

Where Uptime Tiers define the *consequence*, IEC 62443 Security Levels (SLs) define the required *level of protection*. The standard specifies four SLs based on the resources and motivation of the attacker :

- **SL 1:** Protection against *accidental* or *casual* misuse.
- **SL 2:** Protection against *intentional* attack using *simple means* (e.g., "script kiddie").
- **SL 3:** Protection against *intentional* attack using *sophisticated means* (e.g., criminal organization with IACS-specific knowledge).
- **SL 4:** Protection against *intentional* attack using *advanced resources* (e.g., nation-state).

The standard uses three SL designations: SL-T (Target), the desired level; SL-C (Capability), the built-in security of a component; and SL-A (Achieved), the measured security of the final system. This report focuses on defining typical initial **SL-Ts**.

These SLs are measured against Seven Foundational Requirements (FRs) :

1. **FR1:** Identification and Authentication Control (IAC)
2. **FR2:** Use Control (UC)
3. **FR3:** System Integrity (SI)
4. **FR4:** Data Confidentiality (DC)
5. **FR5:** Restricted Data Flow (RDF)
6. **FR6:** Timely Response to Events (TRE)
7. **FR7:** Resource Availability (RA)

A critical distinction exists between IT and OT security priorities. For traditional IT zones, the primary security concerns are FR4 (Data Confidentiality) and FR3 (System Integrity). For data center OT zones (power, cooling), the security model is inverted: the paramount requirements are **FR7 (Resource Availability)** and **FR3 (System Integrity)**. A "secure" OT system is one that *remains available* and cannot be commanded to an unsafe or offline state.

Data Center Functional Asset Groups (The System under Consideration)

Before partitioning the System under Consideration (SuC) into zones, IEC 62443-3-2 requires first identifying all assets. The following table inventories the typical functional "systems" and technologies found in a modern data center, which will form the basis of our zoning model.

Table 2: Data Center Functional Asset Inventory

System Category	Asset Group	Key Technologies & Protocols	Primary Security Concern
Power	Electrical Power Monitoring System (EPMS)	UPS, PDUs, Generators, Switchgear, PLCs, Meters, Modbus, BACnet, SNMP	Availability , Integrity
Cooling	Building Management System (BMS) / HVAC	Chillers, Computer Room Air Conditioners (CRACs), Computer Room Air Handlers (CRAHs), PLCs, Sensors, BACnet, Modbus	Availability , Integrity
Monitoring	Data Center Infrastructure Management (DCIM)	Monitoring Servers, SQL Databases, Analytics Platforms, Web Servers, API Gateways	Integrity, Confidentiality
IT Compute	"White Space" / Compute Zone	Servers, Storage Arrays, Hypervisors, Tenant Racks, Mainframe (if present)	Availability, Confidentiality
Network	Network Operations Center (NOC)	Routers, Core Switches, Firewalls, DNS/DHCP Servers	Availability, Integrity
Security	Physical Security Systems (OT)	Access Control Panels, Biometric Readers, CCTV Servers (NVRs),	Integrity, Availability

System Category	Asset Group	Key Technologies & Protocols	Primary Security Concern
		Physical Access Control Systems (PACS)	

A Typical IEC 62443 Zone & Conduit Model for Data Centers (Baseline Tier II/III)

The Zoning Methodology

Based on the asset inventory and the core IEC 62443 principle—grouping assets with *common security requirements*—a typical, robust segmentation model for a Tier II or Tier III data center can be established. This model goes beyond simplistic 3-zone physical layouts to address the specific cyber-physical risks of a converged environment.

Baseline Zone Definitions

A best-practice baseline model for a Tier II/III facility contains a minimum of seven distinct security zones.

- **Zone 1: Enterprise IT Zone**
 - **Assets:** Corporate user laptops, workstations, file servers, ERP/HR systems, general business applications.
 - **Function:** Standard business operations.
 - **Rationale:** This is the large, "untrusted" corporate network. It has a high density of human users and is a primary target for phishing and common malware. It must be aggressively segregated from all critical data center operations.
- **Zone 2: Data Center IT Zone (Compute Zone)**
 - **Assets:** Server racks, storage arrays, hypervisors, and tenant/customer equipment in a colocation setting.
 - **Function:** Hosts the primary compute workloads and data; this is the core revenue-generating system.
 - **Rationale:** This zone holds the data and applications. It requires high availability and integrity, and must be protected from both the "untrusted" Enterprise Zone and from lateral threats originating in the OT zones.
- **Zone 3: IT/OT Demilitarized Zone (DMZ)**
 - **Assets:** Jump servers (bastion hosts), patch management servers (e.g., WSUS), antivirus update servers, remote access gateways, API gateways.
 - **Function:** A "buffer zone" that serves as the *only* authorized communication path between the Enterprise Zone (Zone 1) and the critical OT/Monitoring zones (Zones 4, 5, 6, 7).
 - **Rationale:** A non-negotiable component of a secure, segmented architecture. All remote access, patch data, and monitoring commands must be brokered and authenticated through this zone.
- **Zone 4: DCIM & Monitoring Zone**
 - **Assets:** DCIM servers, EPMS monitoring applications, BMS monitoring dashboards, and their associated databases.
 - **Function:** Aggregates, analyzes, and visualizes data from all facility OT systems.
 - **Rationale:** This zone is a high-value *reconnaissance target* for an attacker. It has

read-access to all OT systems and provides a complete map of the facility's operations. Compromising it gives an attacker the "keys to the kingdom." It must be segregated from the control devices it monitors to prevent a compromise from propagating.

- **Zone 5: Power Systems Control Zone (EPMS OT)**
 - **Assets:** PLCs, controllers, and smart devices for UPSs, PDUs, generators, and switchgear.
 - **Function:** Manages and *controls* the physical electrical infrastructure.
 - **Rationale:** This is arguably the most critical OT zone. A malicious "write" command to a PLC in this zone could *instantly* cause a facility-wide outage (e.g., "trip all PDUs"), violating the Uptime Tier guarantee.
- **Zone 6: Cooling & Environmental Zone (BMS OT)**
 - **Assets:** PLCs, controllers, and sensors for chillers, CRAHs, CRACs, HVAC, and fire suppression systems.
 - **Function:** Manages and *controls* the physical cooling and life safety infrastructure.
 - **Rationale:** The second most critical OT zone. A compromise here (e.g., "shut down all chillers") would lead to a catastrophic thermal shutdown of the entire IT Compute Zone within minutes. This represents the well-known "HVAC attack vector".
- **Zone 7: Physical Security Zone (OT)**
 - **Assets:** Access control panels, biometric readers, CCTV servers (NVRs), badge systems.
 - **Function:** Controls, monitors, and logs all physical access to the facility, data halls, and racks.
 - **Rationale:** While also an OT system, its security requirements and failure mode are different from power and cooling. A compromise here does not cause an *immediate* IT outage but creates a massive compliance and security breach. Per IEC 62443, assets with different security requirements must be in different zones.

Key Conduits (Logical Groupings of Data Flow)

Conduits are the logical groupings of communication channels that connect zones. All communication between zones must be explicitly defined and enforced, typically by firewalls.

- **C-1-3 (Enterprise-to-DMZ):** Allows authenticated user access (e.g., SSH, RDP) to jump servers in the DMZ.
- **C-3-4 (DMZ-to-Monitoring):** Allows authenticated remote access to the DCIM/BMS dashboards. Allows patch/AV data to flow from DMZ servers to monitoring servers.
- **C-4-5 (Monitoring-to-Power):** Allows *read-only* monitoring protocols (e.g., Modbus/TCP, SNMP) from the DCIM server (Zone 4) to the power PLCs (Zone 5).
- **C-4-6 (Monitoring-to-Cooling):** Allows *read-only* monitoring protocols (e.g., BACnet, SNMP) from the DCIM server (Zone 4) to the cooling PLCs (Zone 6).
- **C-4-7 (Monitoring-to-Phys-Security):** Allows *read-only* access to log data from the physical security systems.
- **C-3-5 / C-3-6 (DMZ-to-OT):** *Highly restricted* conduits that may allow authenticated *write/control* commands (e.g., from an engineer's workstation via the jump server) to the OT zones. This aligns with the IEC 62443 principle of Restricted Data Flow (FR5) , ensuring that monitoring (read) and control (write) are separate data flows.

Table 3: Baseline Zone & Conduit Model (Tier II/III Data Center)

Zone ID	Zone Name	Key Assets	Primary FRs	Key Conduits (To/From)
Z1	Enterprise IT	User PCs,	IAC, UC	C-1-3 (to DMZ),

Zone ID	Zone Name	Key Assets	Primary FRs	Key Conduits (To/From)
		Laptops, File Servers		C-1-2 (to DC IT)
Z2	DC IT (Compute)	Servers, Storage, Racks	RA, SI, DC	C-1-2 (from Enterprise)
Z3	IT/OT DMZ	Jump Servers, Patch Servers	SI, RDF	C-1-3 (from Enterprise), C-3-4, C-3-5, C-3-6
Z4	DCIM & Monitoring	DCIM, EPMS, BMS Servers	SI, DC	C-3-4 (from DMZ), C-4-5, C-4-6, C-4-7 (to OT)
Z5	Power Systems OT	UPS/PDU/Gen PLCs & Controllers	RA, SI	C-4-5 (from Monitoring), C-3-5 (from DMZ)
Z6	Cooling Systems OT	Chiller/CRAC PLCs & Controllers	RA, SI	C-4-6 (from Monitoring), C-3-6 (from DMZ)
Z7	Physical Security OT	Access Control, CCTV	SI, UC	C-4-7 (from Monitoring)

Tier-Based Evolution: How Redundancy *Mandates* Zone Duplication (Tier IV)

The Cyber-Physical Fault Tolerance Mandate

The baseline 7-zone model is sufficient for a Tier III facility, but it is *fatally flawed* for a Tier IV "Fault Tolerant" facility.

A Tier IV design mandates at least two independent, active power and cooling distribution paths (e.g., 'A' and 'B' chains) with *no single points of failure*. The purpose of this 2N redundancy is to withstand any single component failure.

If the *control systems* for both the 'A' chain and 'B' chain (e.g., all power PLCs) are placed in the *same security zone* (e.g., the baseline Zone 5), that zone's network switch or firewall becomes a *common-mode failure point*. A single cyberattack (e.g., ransomware or a malicious command) that compromises this single control zone could simultaneously shut down *both* the 'A' and 'B' paths. This would be a catastrophic, facility-wide outage, proving that the expensive 2N physical redundancy was completely bypassed by a singular cyber-failure.

Therefore, to achieve true cyber-physical fault tolerance, the IEC 62443 zone model *must* be evolved to mirror the 2N physical topology.

The Tier IV Evolved Zone Model

For a Tier IV facility, the baseline model is modified. The critical power and cooling zones are *duplicated* to be fully independent.

- **Zone 5A: 'A' Power Systems Control Zone**
 - **Assets:** All PLCs, controllers, and UPSs associated with the 'A' power chain.
- **Zone 5B: 'B' Power Systems Control Zone**
 - **Assets:** All PLCs, controllers, and UPSs associated with the 'B' power chain.
- **Zone 6A: 'A' Cooling & Environmental Zone**

- **Assets:** All PLCs, controllers, and chillers for the 'A' cooling loop.
- **Zone 6B: 'B' Cooling & Environmental Zone**
 - **Assets:** All PLCs, controllers, and chillers for the 'B' cooling loop.

New Conduit Rules: Direct conduits between Zone 5A and Zone 5B (or 6A and 6B) are *strictly forbidden*. This ensures that a compromise in the 'A' control network cannot propagate to the 'B' control network. All monitoring must be arbitrated via the DCIM/Monitoring Zone (Zone 4), which would have separate, read-only conduits to each of the four independent OT zones (C-4-5A, C-4-5B, C-4-6A, C-4-6B).

Table 4: Tier IV (Fault-Tolerant) Zone Model Enhancements

Baseline Zone (Tier II/III)	Tier IV Evolved Zone(s)	Rationale for Change
Zone 5: Power Systems	Zone 5A: 'A' Power Chain Zone 5B: 'B' Power Chain	Prevents common-mode cyber-failure. A single attack cannot compromise both 2N paths, thus preserving Tier IV fault tolerance.
Zone 6: Cooling Systems	Zone 6A: 'A' Cooling Chain Zone 6B: 'B' Cooling Chain	Prevents common-mode cyber-failure. A single attack cannot compromise both 2N paths, thus preserving Tier IV fault tolerance.

Typical Security Level Targets (SL-Ts) for Data Center Zones

Methodology: Assigning SL-Ts Based on Consequence (Tier)

With the zones defined, SL-Ts can be assigned. As established, the primary input for this decision is the *consequence of compromise*, which is directly expressed by the facility's Uptime Tier.

The second input is the *threat profile*. Research indicates that data center OT is targeted by sophisticated, motivated adversaries (an SL-3 threat), not just SL-2 "script kiddies". This leads to a critical conclusion: While some component vendors may advertise SL-2 certification for power management systems, SL-2 only protects against attacks with *simple means*. For a Tier III or Tier IV facility, where the consequence of an outage is catastrophic, accepting an SL-2 posture for a critical OT zone (like power or cooling) represents an unjustifiable mismatch between risk and protection. The combination of a *high-consequence* asset (Tier III/IV) and a *sophisticated* threat (documented attackers) *mandates* an **SL-T 3** for critical OT zones.

Typical SL-T Assignments per Zone

Based on this methodology, the following initial SL-Ts are typical for a best-practice implementation.

- **Zone 1: Enterprise IT Zone**
 - **Typical SL-T: 2**
 - **Rationale:** This zone is large, complex, and has many human users. It is assumed to be a "dirty" network. The cost to achieve SL-3 across the entire enterprise is prohibitive, so the strategy is *containment* at the boundary (the DMZ).
- **Zone 2: Data Center IT Zone (Compute Zone)**

- **Typical SL-T: 3**
 - *Rationale:* This is the core revenue-generating asset. The consequence of failure is high. It must be protected against sophisticated attacks (SL-3) for Availability, Integrity, and Confidentiality.
- **Zone 3: IT/OT DMZ**
 - **Typical SL-T: 3**
 - *Rationale:* As the "gatekeeper" to the most critical zones, its integrity must be at least as high as the zones it protects. This zone is the primary boundary protection.
- **Zone 4: DCIM & Monitoring Zone**
 - **Typical SL-T: 3**
 - *Rationale:* A high-value target for attacker reconnaissance. A compromise here does not cause an outage, but it provides the "map" for an SL-3 attacker to target the OT zones. Its integrity is paramount.
- **Zone 5: Power Systems Control Zone (and 5A/5B)**
 - **Typical SL-T: 2 (for Tier I/II) | SL-T: 3 (for Tier III/IV)**
 - *Rationale:* This is the *crown jewel* of data center OT. For a Tier III/IV facility, the consequence of failure is catastrophic. The SL-T *must* be 3, specifically focused on **FR7 (Resource Availability)** and **FR3 (System Integrity)**.
- **Zone 6: Cooling & Environmental Zone (and 6A/6B)**
 - **Typical SL-T: 2 (for Tier I/II) | SL-T: 3 (for Tier III/IV)**
 - *Rationale:* Same as Zone 5. The threat of an HVAC-based attack leading to a thermal shutdown is a Tier-violating event. Must be SL-T 3 for **FR7 (Availability)** and **FR3 (Integrity)**.
- **Zone 7: Physical Security Zone**
 - **Typical SL-T: 2**
 - *Rationale:* A compromise here is a severe compliance and security breach but does not cause an *immediate* facility-down event in the same way as power/cooling. SL-T 2 (protection from intentional, simple attack) is a common baseline, though this may be raised to SL-3 based on the data confidentiality needs of the tenants.

Conclusion and Direct Answers to User Questions

Summary of Findings

The analysis reveals a clear, modern methodology for securing data centers using IEC 62443, departing from obsolete models like Purdue.

1. A data center is a converged IT/OT cyber-physical system, and its greatest threat to availability comes from its newly-connected OT (power, cooling) infrastructure.
2. The Uptime Institute Tier standard (criticality) must be used as the primary *consequence analysis* for the IEC 62443-3-2 risk assessment.
3. A Tier IV "Fault Tolerant" (2N) design *mandates* a duplicated 2N cyber-physical zone model for power and cooling to prevent common-mode cyber-failures.
4. The catastrophic consequences of downtime in a Tier III/IV facility, combined with a documented SL-3 (sophisticated) threat profile, *mandates* an **SL-T 3** for critical OT zones, with a primary focus on **FR7 (Resource Availability)**.

Answer 1: "What is the average or typical number of zones in a datacenter?"

There is no single "average" number, as the correct number of zones is a direct *function of the*

data center's criticality (Tier) and security maturity.

- A **low-criticality (Tier I) or non-compliant** data center may have as few as **3-4 zones** (e.g., Enterprise, DC IT, and a single, flat "Facilities/OT" zone). This is a high-risk architecture.
- A **best-practice, IEC 62443-compliant Tier II or Tier III** data center will typically have a minimum of **7 independent zones** (as defined in the baseline model: Enterprise IT, DC IT, IT/OT DMZ, DCIM/Monitoring, Power, Cooling, and Physical Security).
- A **high-criticality, Tier IV "Fault Tolerant"** data center *requires* further segmentation to protect its 2N redundancy. This results in a typical model of **9 or more zones**, as the critical Power (Zone 5) and Cooling (Zone 6) zones are duplicated (Zone 5A/5B, Zone 6A/6B).

Therefore, the typical range for a modern, secure data center is **7 to 9 zones**, with the number increasing with criticality.

Answer 2: "What are the average or typical initial Security level targets for each?"

The typical initial SL-Ts vary by zone function (IT vs. OT) and, most importantly, by the facility's Uptime Tier.

Table 5: Summary of Typical Initial SL-Ts for Data Center Zones

Zone ID	Zone Name	Typical SL-T (Tier I/II)	Typical SL-T (Tier III/IV)	Primary FRs & Rationale
Z1	Enterprise IT	SL-T 2	SL-T 2	(IAC, UC). Assumed "dirty" network. Focus on boundary enforcement, not zone-wide prevention.
Z2	DC IT (Compute)	SL-T 2	SL-T 3	(RA, SI, DC). Core revenue asset. Tier III/IV criticality mandates protection from sophisticated threats.
Z3	IT/OT DMZ	SL-T 2	SL-T 3	(SI, RDF). Boundary must be as secure as the critical zones it protects.
Z4	DCIM/Monitoring	SL-T 2	SL-T 3	(SI, DC). High-value reconnaissance target. Compromise provides the "map" for an SL-3 attack.
Z5	Power Systems OT	SL-T 2	SL-T 3	(RA, SI). Catastrophic

Zone ID	Zone Name	Typical SL-T (Tier I/II)	Typical SL-T (Tier III/IV)	Primary FRs & Rationale
				consequence. Tier III/IV <i>must</i> protect against <i>sophisticated</i> attacks (SL-3).
Z6	Cooling Systems OT	SL-T 2	SL-T 3	(RA, SI). Catastrophic consequence. Tier III/IV <i>must</i> protect against <i>sophisticated</i> attacks (SL-3).
Z7	Physical Security OT	SL-T 2	SL-T 2	(SI, UC). Critical, but not an immediate-down time vector. SL-2 is a common baseline; may be SL-3 if data requires it.

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